

MotorizedBicycle

A Schema.org Type

Defined in the *auto* section.

Thing > **Product** > **Vehicle** > **MotorizedBicycle**

[more...]

A motorized bicycle is a bicycle with an attached motor used to power the vehicle, or to assist with pedaling.

Property	Expected Type	Description
Properties from Vehicle		
	QuantitativeValue	The time needed to accelerate the vehicle from a given start velocity to a given target velocity. Typical unit code(s): SEC for seconds
accelerationTime		Note: There are unfortunately no standard unit codes for seconds/0..100 km/h or seconds/0..60 mph. Simply use "SEC" for seconds and indicate the velocities in the name of the QuantitativeValue, or use valueReference with a QuantitativeValue of 0..60 mph or 0..100 km/h to specify the reference speeds.

Property	Expected Type	Description
bodyType	QualitativeValue or Text or URL	Indicates the design and body style of the vehicle (e.g. station wagon, hatchback, etc.).
callSign	Text	A callsign , as used in broadcasting and radio communications to identify people, radio and TV stations, or vehicles.
cargoVolume	QuantitativeValue	The available volume for cargo or luggage. For automobiles, this is usually the trunk volume. Typical unit code(s): LTR for liters, FTQ for cubic foot/feet Note: You can use minValue and maxValue to indicate ranges.
dateVehicleFirstRegistered	Date	The date of the first registration of the vehicle with the respective public authorities.
driveWheelConfiguration	DriveWheelConfigurationValue or Text	The drive wheel configuration, i.e. which roadwheels will receive torque from the vehicle's engine via the drivetrain.
emissionsCO2	Number	The CO2 emissions in g/km. When used in combination with a QuantitativeValue, put "g/km" into the unitText property of that value, since there is no UN/CEFACT Common Code for "g/km".
fuelCapacity	QuantitativeValue	The capacity of the fuel tank or in the case of electric cars, the battery. If there are multiple components for storage, this should indicate the total of all storage of the same type.

Property	Expected Type	Description
		Typical unit code(s): LTR for liters, GLL of US gallons, GLI for UK / imperial gallons, AMH for ampere-hours (for electrical vehicles).
fuelConsumption	QuantitativeValue	The amount of fuel consumed for traveling a particular distance or temporal duration with the given vehicle (e.g. liters per 100 km). - Note 1: There are unfortunately no standard unit codes for liters per 100 km. Use unitText to indicate the unit of measurement, e.g. L/100 km. - Note 2: There are two ways of indicating the fuel consumption, fuelConsumption (e.g. 8 liters per 100 km) and fuelEfficiency (e.g. 30 miles per gallon). They are reciprocal. - Note 3: Often, the absolute value is useful only when related to driving speed ("at 80 km/h") or usage pattern ("city traffic"). You can use valueReference to link the value for the fuel consumption to another value.
fuelEfficiency	QuantitativeValue	The distance traveled per unit of fuel used; most commonly miles per gallon (mpg) or kilometers per liter (km/L). Note 1: There are unfortunately no standard unit codes for miles per gallon or kilometers per liter. Use unitText to indicate the unit of measurement, e.g. mpg or km/L.

Property	Expected Type	Description
		- Note 2: There are two ways of indicating the fuel consumption, fuelConsumption (e.g. 8 liters per 100 km) and fuelEfficiency (e.g. 30 miles per gallon). They are reciprocal. - Note 3: Often, the absolute value is useful only when related to driving speed ("at 80 km/h") or usage pattern ("city traffic"). You can use valueReference to link the value for the fuel economy to another value.
fuelType	QualitativeValue or Text or URL	The type of fuel suitable for the engine or engines of the vehicle. If the vehicle has only one engine, this property can be attached directly to the vehicle.
knownVehicleDamages	Text	A textual description of known damages, both repaired and unrepaired.
meetsEmissionStandard	QualitativeValue or Text or URL	Indicates that the vehicle meets the respective emission standard.
mileageFromOdometer	QuantitativeValue	The total distance travelled by the particular vehicle since its initial production, as read from its odometer. Typical unit code(s): KMT for kilometers, SMI for statute miles.
modelDate	Date	The release date of a vehicle model (often used to differentiate versions of the same make and model).
numberOfAirbags	Number or Text	The number or type of airbags in the vehicle.

Property	Expected Type	Description
numberOfAxles	Number or QuantitativeValue	The number of axles. Typical unit code(s): C62.
numberOfDoors	Number or QuantitativeValue	The number of doors. Typical unit code(s): C62.
numberOfForwardGears	Number or QuantitativeValue	The total number of forward gears available for the transmission system of the vehicle. Typical unit code(s): C62.
numberOfPreviousOwners	Number or QuantitativeValue	The number of owners of the vehicle, including the current one. Typical unit code(s): C62.
payload	QuantitativeValue	The permitted weight of passengers and cargo, EXCLUDING the weight of the empty vehicle. Typical unit code(s): KGM for kilogram, LBR for pound - Note 1: Many databases specify the permitted TOTAL weight instead, which is the sum of weight and payload - Note 2: You can indicate additional information in the name of the QuantitativeValue node.

Property	Expected Type	Description
		- Note 3: You may also link to a QualitativeValue node that provides additional information using valueReference. - Note 4: Note that you can use minValue and maxValue to indicate ranges.
productionDate	Date	The date of production of the item, e.g. vehicle.
purchaseDate	Date	The date the item, e.g. vehicle, was purchased by the current owner.
seatingCapacity	Number or QuantitativeValue	The number of persons that can be seated (e.g. in a vehicle), both in terms of the physical space available, and in terms of limitations set by law. Typical unit code(s): C62 for persons.

Property	Expected Type	Description
speed	QuantitativeValue	The speed range of the vehicle. If the vehicle is powered by an engine, the upper limit of the speed range (indicated by maxValue) should be the maximum speed achievable under regular conditions. Typical unit code(s): KMH for km/h, HM for mile per hour (0.447 04 m/s), KNT for knot *Note 1: Use minValue and maxValue to indicate the range. Typically, the minimal value is zero. * Note 2: There are many different ways of measuring the speed range. You can link to information about how the given value has been determined using the valueReference property.
	SteeringPositionValue	The position of the steering wheel or similar device (mostly for cars).
steeringPosition		
tongueWeight	QuantitativeValue	The permitted vertical load (TWR) of a trailer attached to the vehicle. Also referred to as Tongue Load Rating (TLR) or Vertical Load Rating (VLR). Typical unit code(s): KGM for kilogram, LBR for pound - Note 1: You can indicate additional information in the name of the QuantitativeValue node. - Note 2: You may also link to a QualitativeValue node that provides additional information using valueReference.

Property	Expected Type	Description
		Note 3: Note that you can use minValue and maxValue to indicate ranges.
trailerWeight	QuantitativeValue	The permitted weight of a trailer attached to the vehicle. Typical unit code(s): KGM for kilogram, LBR for pound * Note 1: You can indicate additional information in the name of the QuantitativeValue node. * Note 2: You may also link to a QualitativeValue node that provides additional information using valueReference. * Note 3: Note that you can use minValue and maxValue to indicate ranges.
vehicleConfiguration	Text	A short text indicating the configuration of the vehicle, e.g. '5dr hatchback ST 2.5 MT 225 hp' or 'limited edition'.
vehicleEngine	EngineSpecification	Information about the engine or engines of the vehicle.
vehicleIdentificationNumber	Text	The Vehicle Identification Number (VIN) is a unique serial number used by the automotive industry to identify individual motor vehicles.
vehicleInteriorColor	Text	The color or color combination of the interior of the vehicle.
vehicleInteriorType	Text	The type or material of the interior of the vehicle (e.g. synthetic fabric, leather, wood, etc.). While most interior types are characterized by the material used, an interior type can also be based on vehicle usage or target audience.
vehicleModelDate	Date	The release date of a vehicle model (often used to differentiate versions of the same make and model).

Property	Expected Type	Description
vehicleSeatingCapacity	Number or QuantitativeValue	The number of passengers that can be seated in the vehicle, both in terms of the physical space available, and in terms of limitations set by law. Typical unit code(s): C62 for persons.
vehicleSpecialUsage	CarUsageType or Text	Indicates whether the vehicle has been used for special purposes, like commercial rental, driving school, or as a taxi. The legislation in many countries requires this information to be revealed when offering a car for sale.
vehicleTransmission	QualitativeValue or Text or URL	The type of component used for transmitting the power from a rotating power source to the wheels or other relevant component(s) ("gearbox" for cars).
weightTotal	QuantitativeValue	The permitted total weight of the loaded vehicle, including passengers and cargo and the weight of the empty vehicle. Typical unit code(s): KGM for kilogram, LBR for pound - Note 1: You can indicate additional information in the name of the QuantitativeValue node. - Note 2: You may also link to a QualitativeValue node that provides additional information using valueReference. - Note 3: Note that you can use minValue and maxValue to indicate ranges.

Property	Expected Type	Description
wheelbase	QuantitativeValue	The distance between the centers of the front and rear wheels. Typical unit code(s): CMT for centimeters, MTR for meters, INH for inches, FOT for foot/feet.
Properties from Product		
additionalProperty	PropertyValue	A property-value pair representing an additional characteristic of the entity, e.g. a product feature or another characteristic for which there is no matching property in schema.org. Note: Publishers should be aware that applications designed to use specific schema.org properties (e.g. https://schema.org/width, https://schema.org/color, https://schema.org/gtin13, ...) will typically expect such data to be provided using those properties, rather than using the generic property/value mechanism.
aggregateRating	AggregateRating	The overall rating, based on a collection of reviews or ratings, of the item.
asin	Text or URL	An Amazon Standard Identification Number (ASIN) is a 10-character alphanumeric unique identifier assigned by Amazon.com and its partners for product identification within the Amazon organization (summary from Wikipedia's article). Note also that this is a definition for how to include ASINs in Schema.org data, and not a definition of ASINs in general - see documentation from Amazon for authoritative details. ASINs

Property	Expected Type	Description
		are most commonly encoded as text strings, but the [asin] property supports URL/URI as potential values too.
audience	Audience	An intended audience, i.e. a group for whom something was created. Supersedes serviceAudience .
award	Text	An award won by or for this item. Supersedes awards .
brand	Brand or Organization	The brand(s) associated with a product or service, or the brand(s) maintained by an organization or business person.
category	CategoryCode or PhysicalActivityCategory or Text or Thing or URL	A category for the item. Greater signs or slashes can be used to informally indicate a category hierarchy.
color	Text	The color of the product.
colorSwatch	ImageObject or URL	A color swatch image, visualizing the color of a Product . Should match the textual description specified in the color property. This can be a URL or a fully described ImageObject.
countryOfAssembly	Text	The place where the product was assembled.
countryOfLastProcessing	Text	The place where the item (typically Product) was last processed and tested before importation.
countryOfOrigin	Country	The country of origin of something, including products as well as creative works such as movie and TV content.

Property	Expected Type	Description
		In the case of TV and movie, this would be the country of the principle offices of the production company or individual responsible for the movie. For other kinds of CreativeWork it is difficult to provide fully general guidance, and properties such as contentLocation and locationCreated may be more applicable. In the case of products, the country of origin of the product. The exact interpretation of this may vary by context and product type, and cannot be fully enumerated here.
	Distance or QuantitativeValue	The depth of the item.
funding	Grant	A Grant that directly or indirectly provide funding or sponsorship for this item. See also ownershipFundingInfo. Inverse property: fundedItem
gtin	Text or URL	A Global Trade Item Number (GTIN). GTINs identify trade items, including products and services, using numeric identification codes. A correct gtin value should be a valid GTIN, which means that it should be an all-numeric string of either 8, 12, 13 or 14 digits, or a "GS1 Digital Link" URL based on such a string. The numeric component should also have a valid GS1 check digit and meet the other rules for valid GTINs. See also GS1's GTIN Summary and Wikipedia for more details. Left-padding of the gtin values is not required or encouraged. The gtin property generalizes the earlier gtin8, gtin12, gtin13, and gtin14 properties.

Property	Expected Type	Description
		The GS1 digital link specifications expresses GTINs as URLs (URIs, IRIs, etc.). Digital Links should be populated into the hasGS1DigitalLink attribute. Note also that this is a definition for how to include GTINs in Schema.org data, and not a definition of GTINs in general - see the GS1 documentation for authoritative details.
gtin12	Text	The GTIN-12 code of the product, or the product to which the offer refers. The GTIN-12 is the 12-digit GS1 Identification Key composed of a U.P.C. Company Prefix, Item Reference, and Check Digit used to identify trade items. See GS1 GTIN Summary for more details.
gtin13	Text	The GTIN-13 code of the product, or the product to which the offer refers. This is equivalent to 13-digit ISBN codes and EAN UCC-13. Former 12-digit UPC codes can be converted into a GTIN-13 code by simply adding a preceding zero. See GS1 GTIN Summary for more details.
gtin14	Text	The GTIN-14 code of the product, or the product to which the offer refers. See GS1 GTIN Summary for more details.
gtin8	Text	The GTIN-8 code of the product, or the product to which the offer refers. This code is also known as EAN/UCC-8 or 8-digit EAN. See GS1 GTIN Summary for more details.
hasAdultConsideration	AdultOrientedEnumeration	Used to tag an item to be intended or suitable for consumption or use by adults only.

Property	Expected Type	Description
hasCertification	Certification	Certification information about a product, organization, service, place, or person.
hasEnergyConsumptionDetails	EnergyConsumptionDetails	Defines the energy efficiency Category (also known as "class" or "rating") for a product according to an international energy efficiency standard.
hasGS1DigitalLink	URL	The GS1 digital link associated with the object. This URL should conform to the particular requirements of digital links. The link should only contain the Application Identifiers (AIs) that are relevant for the entity being annotated, for instance a Product or an Organization, and for the correct granularity. In particular, for products: • A Digital Link that contains a serial number (AI 21) should only be present on instances of IndividualProduct • A Digital Link that contains a lot number (AI 10) should be annotated as SomeProduct if only products from that lot are sold, or IndividualProduct if there is only a specific product. • A Digital Link that contains a global model number (AI 8013) should be attached to a Product or a ProductModel. Other item types should be adapted similarly.
hasMeasurement	QuantitativeValue	A measurement of an item, For example, the inseam of pants, the wheel size of a bicycle, the gauge of a screw, or the carbon footprint measured for certification by an authority. Usually an

Property	Expected Type	Description
		exact measurement, but can also be a range of measurements for adjustable products, for example belts and ski bindings.
hasMerchantReturnPolicy	MerchantReturnPolicy	Specifies a MerchantReturnPolicy that may be applicable. Supersedes hasProductReturnPolicy .
height	Distance or QuantitativeValue	The height of the item.
inProductGroupWithID	Text	Indicates the productGroupID for a ProductGroup that this product isVariantOf .
isAccessoryOrSparePartFor	Product	A pointer to another product (or multiple products) for which this product is an accessory or spare part.
isConsumableFor	Product	A pointer to another product (or multiple products) for which this product is a consumable.
isFamilyFriendly	Boolean	Indicates whether this content is family friendly.
isRelatedTo	Product or Service	A pointer to another, somehow related product (or multiple products).
isSimilarTo	Product or Service	A pointer to another, functionally similar product (or multiple products).
isVariantOf	ProductGroup or ProductModel	Indicates the kind of product that this is a variant of. In the case of ProductModel , this is a pointer (from a ProductModel) to a base product from which this product is a variant. It is safe to infer that the variant inherits all product features from the base model, unless defined locally. This is

Property	Expected Type	Description
		not transitive. In the case of a ProductGroup, the group description also serves as a template, representing a set of Products that vary on explicitly defined, specific dimensions only (so it defines both a set of variants, as well as which values distinguish amongst those variants). When used with ProductGroup, this property can apply to any Product included in the group. Inverse property: hasVariant
itemCondition	OfferItemCondition	A predefined value from OfferItemCondition specifying the condition of the product or service, or the products or services included in the offer. Also used for product return policies to specify the condition of products accepted for returns.
keywords	DefinedTerm or Text or URL	Keywords or tags used to describe some item. Multiple textual entries in a keywords list are typically delimited by commas, or by repeating the property.
logo	ImageObject or URL	An associated logo.
manufacturer	Organization	The manufacturer of the product.
material	Product or Text or URL	A material that something is made from, e.g. leather, wool, cotton, paper.
mobileUrl	Text	The mobileUrl property is provided for specific situations in which data consumers need to determine whether one of several provided URLs is a dedicated 'mobile site'.

Property	Expected Type	Description
		To discourage over-use, and reflecting initial usecases, the property is expected only on Product and Offer, rather than Thing. The general trend in web technology is towards responsive design in which content can be flexibly adapted to a wide range of browsing environments. Pages and sites referenced with the long-established url property should ideally also be usable on a wide variety of devices, including mobile phones. In most cases, it would be pointless and counter productive to attempt to update all url markup to use mobileUrl for more mobile-oriented pages. The property is intended for the case when items (primarily Product and Offer) have extra URLs hosted on an additional "mobile site" alongside the main one. It should not be taken as an endorsement of this publication style.
model	ProductModel or Text	The model of the product. Use with the URL of a ProductModel or a textual representation of the model identifier. The URL of the ProductModel can be from an external source. It is recommended to additionally provide strong product identifiers via the gtin8/gtin13/gtin14 and mpn properties.
mpn	Text	The Manufacturer Part Number (MPN) of the product, or the product to which the offer refers.
negativeNotes	ItemList or ListItem or Text or WebContent	Provides negative considerations regarding something, most typically in pro/con lists for reviews (alongside positiveNotes). For symmetry In the case of a Review, the property describes the itemReviewed from the perspective of the review; in the case

Property	Expected Type	Description
		of a Product, the product itself is being described. Since product descriptions tend to emphasise positive claims, it may be relatively unusual to find negativeNotes used in this way. Nevertheless for the sake of symmetry, negativeNotes can be used on Product. The property values can be expressed either as unstructured text (repeated as necessary), or if ordered, as a list (in which case the most negative is at the beginning of the list).
nsn	Text	Indicates the NATO stock number (nsn) of a Product .
offers	Demand or Offer	An offer to provide this item—for example, an offer to sell a product, rent the DVD of a movie, perform a service, or give away tickets to an event. Use businessFunction to indicate the kind of transaction offered, i.e. sell, lease, etc. This property can also be used to describe a Demand. While this property is listed as expected on a number of common types, it can be used in others. In that case, using a second type, such as Product or a subtype of Product, can clarify the nature of the offer. Inverse property: itemOffered
pattern	DefinedTerm or Text	A pattern that something has, for example 'polka dot', 'striped', 'Canadian flag'. Values are typically expressed as text, although links to controlled value schemes are also supported.
positiveNotes	ItemList or ListItem or	Provides positive considerations regarding something, for example product highlights or (alongside negativeNotes) pro/con lists for reviews.

Property	Expected Type	Description
	Text or WebContent	In the case of a Review, the property describes the itemReviewed from the perspective of the review; in the case of a Product, the product itself is being described. The property values can be expressed either as unstructured text (repeated as necessary), or if ordered, as a list (in which case the most positive is at the beginning of the list).
productID	Text	The product identifier, such as ISBN. For example: meta itemprop="productID" content="isbn:123-456-789".
productionDate	Date	The date of production of the item, e.g. vehicle.
purchaseDate	Date	The date the item, e.g. vehicle, was purchased by the current owner.
releaseDate	Date	The release date of a product or product model. This can be used to distinguish the exact variant of a product.
review	Review	A review of the item. Supersedes reviews .
size	DefinedTerm or QuantitativeValue or SizeSpecification or Text	A standardized size of a product or creative work, specified either through a simple textual string (for example 'XL', '32Wx34L'), a QuantitativeValue with a unitCode, or a comprehensive and structured SizeSpecification ; in other cases, the width , height , depth and weight properties may be more applicable.
sku	Text	The Stock Keeping Unit (SKU), i.e. a merchant-specific identifier for a product or service, or the product to which the

Property	Expected Type	Description
		offer refers.
slogan	Text	A slogan or motto associated with the item.
weight	QuantitativeValue	The weight of the product or person.
width	Distance or QuantitativeValue	The width of the item.
Properties from Thing		
additionalType	Text or URL	An additional type for the item, typically used for adding more specific types from external vocabularies in microdata syntax. This is a relationship between something and a class that the thing is in. Typically the value is a URI-identified RDF class, and in this case corresponds to the use of rdf:type in RDF. Text values can be used sparingly, for cases where useful information can be added without their being an appropriate schema to reference. In the case of text values, the class label should follow the schema.org style guide .
alternateName	Text	An alias for the item.
description	Text or TextObject	A description of the item.
disambiguatingDescription	Text	A sub property of description. A short description of the item used to disambiguate from other, similar items. Information from other properties (in particular, name) may be necessary for the description to be useful for disambiguation.

Property	Expected Type	Description
identifier	PropertyValue or Text or URL	The identifier property represents any kind of identifier for any kind of Thing , such as ISBNs, GTIN codes, UUIDs etc. Schema.org provides dedicated properties for representing many of these, either as textual strings or as URL (URI) links. See background notes for more details.
image	ImageObject or URL	An image of the item. This can be a URL or a fully described ImageObject .
mainEntityOfPage	CreativeWork or URL	Indicates a page (or other CreativeWork) for which this thing is the main entity being described. See background notes for details. Inverse property: mainEntity
name	Text	The name of the item.
potentialAction	Action	Indicates a potential Action, which describes an idealized action in which this thing would play an 'object' role.
sameAs	URL	URL of a reference Web page that unambiguously indicates the item's identity. E.g. the URL of the item's Wikipedia page, Wikidata entry, or official website.
subjectOf	CreativeWork or Event	A CreativeWork or Event about this Thing. Inverse property: about
url	URL	URL of the item.

Acknowledgements

Automotive Ontology Working Group

This element is based on the work of the Automotive Ontology Working Group, see www.automotive-ontology.org for details. Many class and property definitions are inspired by or based on abstracts from Wikipedia, the free encyclopedia.

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